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4. Tree

1. Title

Binary Tree Representation using linked Representation in c++.

2. problem statement.

Write a c++ program to create a binary tree using linked representation and display the data, left child and right child of every node.

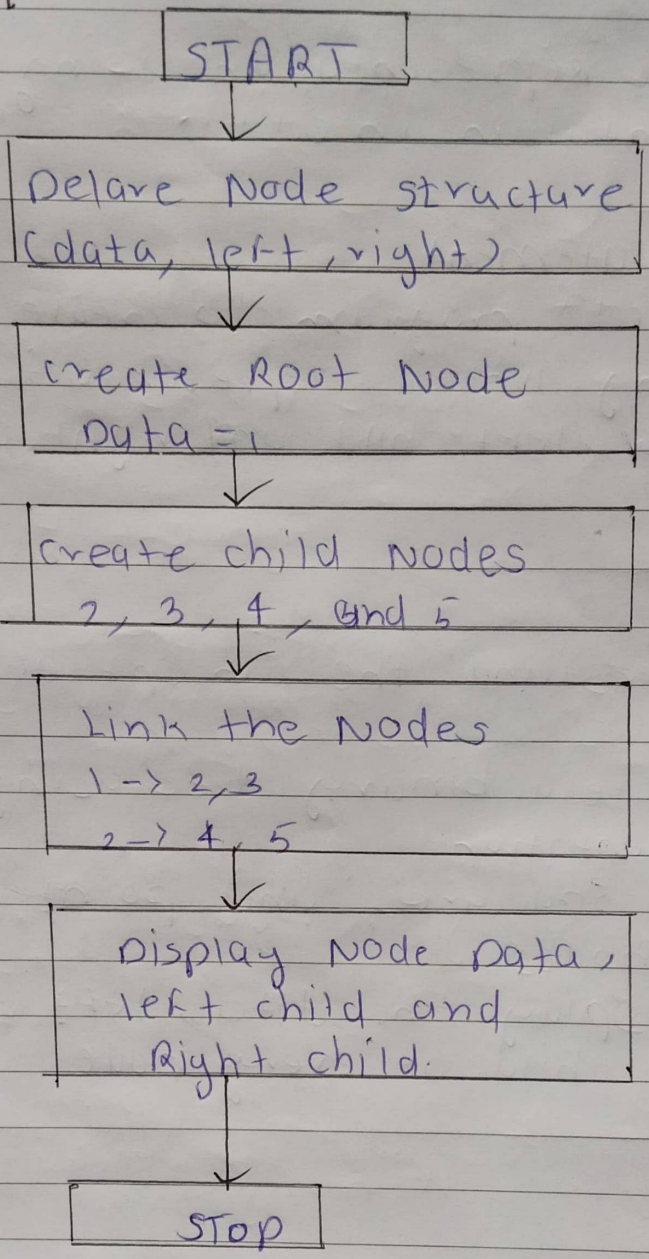
3. objective

1. To understand the concept of a binary tree
2. To create a binary tree using linked representation.
3. To use structures and pointers to create tree nodes.
4. To display the data, left child and right child of every node.

4. Algorithm

1. Start
2. Define a structure Node containing data, a left pointer, and a right pointer.
3. create a function to create a new node.
4. create the root node with data 1.
5. create the left and right children of the root with data 2 and 3.
6. create the left and right children of node 2 with data 4 and 5.
7. Display the data, left child, and right child of every node.
8. Stop

5. Flowchart:-


START
Delare Node structure (data, left, right)
create Root Node Data = 1
create child Nodes 2, 3, 4, and 5
Link the Nodes
1 -> 2, 3
2 -> 4, 5
Display Node Data, left child and Right child.
STOP

```
#include <iostream>
using namespace std;

struct Node {
    int data;
    Node* left;
    Node* right;
};

Node* createNode(int value) {
    Node* newNode = new Node;
    newNode->data = value;
    newNode->left = NULL;
    newNode->right = NULL;
    return newNode;
}

void display(Node* root) {
    if (root == NULL)
        return;

    cout << "Data: " << root->data;

    cout << ", Left Child: ";
    if (root->left != NULL)
        cout << root->left->data;
    else
        cout << "NULL";
```

```
    cout << ", Right Child: ";
    if (root->right != NULL)
        cout << root->right->data;
    else
        cout << "NULL";

    cout << endl;

    display(root->left);
    display(root->right);
}

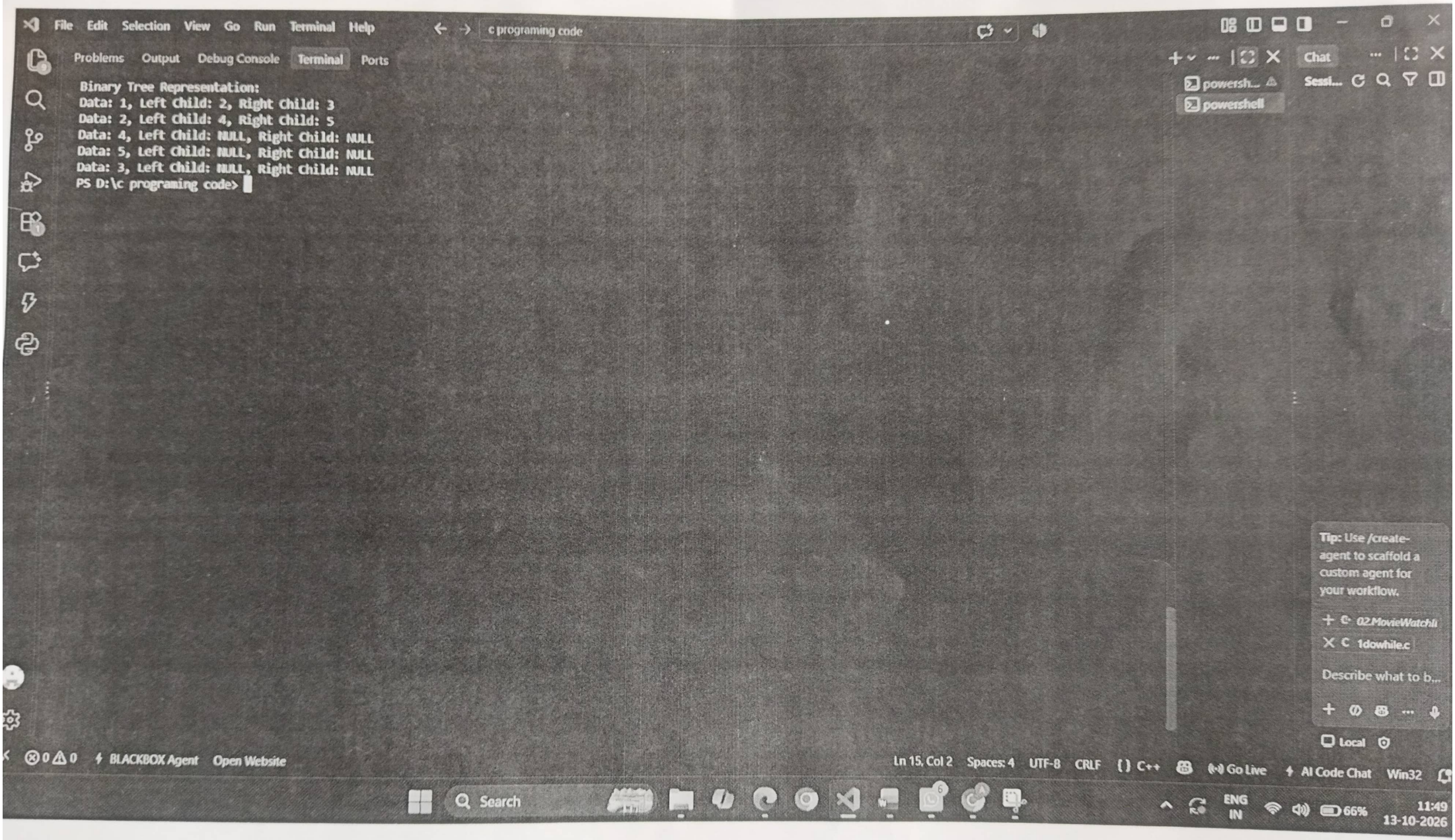
int main() {
    Node* root = createNode(1);

    root->left = createNode(2);
    root->right = createNode(3);

    root->left->left = createNode(4);
    root->left->right = createNode(5);

    cout << "Binary Tree Representation:" << endl;
    display(root);

    return 0;
}
```


File Edit Selection View Go Run Terminal Help
c programing code
Problems Output Debug Console Terminal Ports
Binary Tree Representation:
Data: 1, Left Child: 2, Right Child: 3
Data: 2, Left Child: 4, Right Child: 5
Data: 4, Left Child: NULL, Right Child: NULL
Data: 5, Left Child: NULL, Right Child: NULL
Data: 3, Left Child: NULL, Right Child: NULL
PS D:\c programing code>
Chat
powersh...
powershell
Tip: Use /create-agent to scaffold a custom agent for your workflow.
02.MovieWatchli
1dowhile.c
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Ln 15, Col 2
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